Personal portfolio

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Team	**Team 08**
Build role	**Developer**
Handbook for	**Owners of Dutch family-owned machine builders like Giesen Roasting Solutions**
Entries	**Weeks 1 to 5, September and October 2026**

THE FIVE WEEKS AT A GLANCE

	WEEK THEME	WHAT THE TEAM PUBLISHED	WHAT MY BUILD ROLE PRODUCED
1	**Problem analysis, target audience, research proposal** 7 September 2026	First research proposal and PRD, with Koffiebranderij Peeze as our target SME	Made the PRD's technical side and constraints specific enough to build from
2	**Geopolitics & vendor models** 14 September 2026	Handbook page on geopolitics and vendor models, written for Giesen	Moved our target to Giesen; the technical blueprint; the vendor-ownership research
3	**Trust & AI safety** 21 September 2026	Handbook page on trust and AI safety	Knowledge architecture, first decision-log entry, first running tool in Claude Code
4	**SME knowledge gaps** 28 September 2026	Knowledge-gaps page in progress; the mini-CBI with the lecturers	The live demo of the tool; GitHub as the first subtask of the build plan
5	**AI use cases at SMEs** 5 October 2026	Pages on SME knowledge gaps and AI use cases; our postcard from 2035	An LLM wiki with a firewall and quotation check; a stricter `/check`; 41 claims tested

Week 1

Problem analysis, target audience, research proposal

Session Mon 7 September 2026

WHAT I CONTRIBUTED

I picked Koffiebranderij Peeze, a coffee roaster in Arnhem and one of HAN's partner companies, as our target SME instead of working from a generic invented sector. Once we had a first pass at the research proposal and PRD, I pushed back when parts of it were built on assumptions that didn't actually hold up for Peeze: unverified claims and generic buzzwords standing in for Peeze's real situation. I used Claude to fact-check the company (ownership, sustainability positioning, what they actually do beyond roasting) and rework both drafts so they're grounded in what's true rather than what sounds impressive. I drove the target choice and what stayed in the final draft; the AI helped me verify and structure it faster than I'd have managed alone this week.

WHAT MY BUILD ROLE DID FIRST

I'm the developer. Nothing gets built yet this week, so my first job was making sure the PRD's technical side (what it must let a user do, the Dutch translation requirement, manual fallbacks, agentic CLI versus hand-coded scripts) was specific enough to actually build from rather than aspirational bullet points. I treated the constraints section as the thing I'll be held to from week 2 on, so I made sure it said what I can realistically deliver.

WHAT I DON'T UNDERSTAND YET

I don't yet know how much of the agentic CLI I'm expected to build from scratch versus wiring together existing tools, or what "direct an agentic CLI, don't hand-code" concretely means in practice for something like Dutch-to-English source translation. I also don't know how much our working decision (predictive maintenance for Peeze's leased machines) survives contact with the actual interview.

Week 2

Geopolitics & vendor models

Session Mon 14 September 2026

WHAT DID THE TEAM PUBLISH?

Our page on geopolitics and vendor models, written for Giesen Roasting Solutions, a family-owned precision manufacturer in Ulft/Dinxperlo. It argues something more mixed than the usual 'no European option' story: the foundational production-scheduling layer does have a genuine EU option at SME pricing (an Estonian company), but the AI-assisted digital-twin layer, the kind HAN's own Lean QRM Centrum is already building with Giesen, has no commercial equivalent at this scale outside a handful of venture-backed American platforms. The page ends with what that split means in practice for a firm Giesen's size.

WHAT DID I CONTRIBUTE TO IT?

I drove the change of target company this week, and I want to be specific about why rather than just note that it happened. Our week 1 choice, Koffiebranderij Peeze, ran into a real problem: whether it still counted as an independent SME depended on facts about its owner, Coffee Growth Group, that only an interview could settle, which made our whole research proposal contingent on a visit we had not had yet. Rather than pick a different company by guesswork, I searched HAN's own published research pages nationally for a company with a documented, verifiable partnership, not an inferred one. I found that Giesen is one of six companies working directly with HAN's Lean QRM Centrum on a digital-twin fieldlab, sourced from HAN's own site rather than from the company's marketing. I verified Giesen's history, size and the 2025 rebrand independently before we built the proposal around it.

For the handbook page specifically, I did the vendor-ownership research the argument rests on: tracing the SME-accessible production-scheduling tools to their owners and headquarters, which is where the Estonia-versus-US split came from. That took the page somewhere I did not expect going in. I assumed we would find the same 'no EU option' pattern as our earlier draft, and the evidence did not support that this time, so the page says something more careful instead.

WHAT DID THE GATES ASK, AND WHAT DID I CHANGE?

The Socratic tutor's five questions on our PRD were largely the same shape as before, but this time I could give sharper answers because the supervisor had separately pushed us on a related point: that 'we will check quality' is not a criterion without a published standard, a measurement method and a threshold. Building that standard first changed how I answered the metrics question: instead of inventing new numbers, I pointed the PRD at the standard's own thresholds, so there is one definition of quality enforced in two places instead of two definitions that could drift apart.

The prioritization question changed my actual view, not just my answer. I had been treating source retrieval as the obvious first priority, since nothing works without it. Working through the measurability point made the stronger case for the check step: a platform that retrieves fine but

cannot verify anything cannot tell you whether it worked, which is the entire thing being tested. I now treat Add and Check as equally first-priority, and said so in the PRD rather than leaving my original answer standing.

WHAT DID MY BUILD ROLE PRODUCE?

The technical blueprint: seven parts each with one owning role, an orchestrator for control, and five file contracts. The seam I care about most is translate to draft as claims.md, which keeps Giesen's own production vocabulary and HAN's Dutch-language fieldlab reporting auditable rather than trusted blind. The line is also more specific than last week's version: not just interview material, but any production, scheduling or proprietary fieldlab data Giesen or HAN shares in confidence.

I left two things open rather than filled: what happens when a part has a real source but has misread it, which our check step does not catch; and whether HAN's own unpublished fieldlab findings need a publishability check before we cite them, which the PRD raised and the blueprint enforces without resolving. What I do not have yet is a running pipeline, and the first thing I need to test in week 3 is whether the model actually holds a long Dutch technical document without dropping detail from the middle of it, since check assumes it does and I have not verified that.

Week 3

Trust & AI safety

Session Mon 21 September 2026

WHAT DID THE TEAM PUBLISH?

Our page on trust and AI safety, written for a Dutch family-owned machine builder like Giesen. It argues that an owner should not try to decide whether AI is safe in general. For each use, decide three things: which data may leave the building, whether the tool may only advise or may also act, and how staff tell you they use it. It ends with a small example that can be undone: a late-order warning list that advises the planner and never changes the plan. The page describes the type of firm and does not name Giesen, because we have no written consent yet.

WHAT DID I CONTRIBUTE TO IT?

Less text than last week, and most of what I did was rejecting things. I used Claude to rework our draft, and the first reworked version was wrong in a way that was easy to miss. It had taken the example from the assignment, a region and a target audience the lecturers use only to illustrate the task, and written it into our documents as if it were our own research. It read well. I caught it because the audience on the page was no longer the firm we had chosen, and I had the page and the proposal rebuilt around our own sector. What I can back up is that decision, and the rule that came out of it: the assignment text is a guideline for us and never a source.

WHAT DID THE GATES ASK, AND WHAT DID I CHANGE?

When we wrote the first version of the architecture on 23 September, the tutor's questions on our blueprint had not reached us. What we had was a letter about the search strategy and rubric in our research proposal, which we answered in the proposal itself. So we put the blueprint next to the PRD ourselves and found four places where they no longer matched: the PRD tracked five measures and the blueprint computed two; long sources had no step; a check on cost claims had no home; and no part owned the translation of Dutch sources.

At first we patched each one, which made the design bigger. The redesign I describe below removed most of them instead. The PRD now has five measures, and each names who checks it. Translation has one home: the appraise step keeps the Dutch sentence as the evidence and puts the English beside it. So the change that mattered most this week did not come from a gate question. It came from reading our own two documents side by side.

WHAT DID MY BUILD ROLE PRODUCE?

The knowledge architecture, the first entry of our decision log, and the first version of the tool that actually runs.

Our architecture of 23 September had seven named components and an Evidence Store with three tables. It was more than four people can build in the weeks we have, and none of it ran. I went back to the concepts from class (LLM, skills, tools, context, memory) and cut the design

down to what those need: six skills that a person types (`/start`, `/search`, `/appraise`, `/draft`, `/check`, `/save`), our own documents as context, and two plain files. One is an evidence log that keeps the exact source sentence for every claim. The other is a session log, which is the memory.

I first designed this as a Claude Project. Checked against the constraint in our PRD, “direct an agentic CLI”, it did not hold: a chat project is not a CLI, it cannot write its own log files, and it cannot be blocked from a folder. So I moved it to Claude Code. The whole agent is now one folder: an instruction file, six skill files, and a settings file that blocks a protected or interviews folder inside the project. I wrote no code, and I did not write those files by hand either. I directed Claude to draft them, set the folder up on my laptop and tested it. The test that counts worked: `/save` wrote the session log, and a new session answered `/start` with what had just been done.

What is still open: the pages of weeks 2 and 3 were written before the tool existed, so it has not yet taken a page from search to check. And the test I set myself in week 2, whether the model can hold a long Dutch document, has become a different question. The tool now copies the exact sentence for each claim into the evidence log and checks the draft against that row. That is a design answer, and it still has to be shown on a long Dutch source.

Week 4

SME knowledge gaps

Session Mon 28 September 2026

WHAT DID THE TEAM PUBLISH?

Our page for this week is on SME knowledge gaps. It covers sub-questions E3 and I3 of our proposal: what Dutch manufacturing SMEs report as their main barriers to using AI, and what a machine builder knows about its own delivery-date performance and planning data. On the day I write this, Friday 2 October, the page is still being written and is not on the site yet. No company visit is booked either, so the internal question will be answered from published cases of comparable manufacturers, labelled as such, which is the fallback in our proposal.

WHAT DID I CONTRIBUTE TO IT?

My main contribution this week is the tool, not the text. This page is the first one that can go through the tool from `/search` to `/check`, and making that possible was my part: the folder, the skills and the two logs it works from. That is a smaller contribution to the page itself than the vendor research I did in week 2, and I would rather say so than claim sentences on a page that is not finished.

WHAT DID THE GATES ASK, AND WHAT DID I CHANGE?

This week the gate was the mini-CBI on Monday 28 September: five questions from the lecturers about our tool, our page and our interview guide.

They asked for a demo, and where the tool still falls short. The demo ran live from our project folder: `/start`, a search in English and Dutch, a saved session, and a new session that remembered it. The honest shortfall is that the page we could show next to it had not been made by the tool.

They asked what makes it agentic and which element had the greatest impact. For me that is context. The skills are short because the instruction file loads our research proposal at the start of every session, so `/search` already knows the sub-questions and `/check` already knows our seven criteria. I saw the other side of this when I moved to Claude Code: the documents in the folder still described the old set-up and had to be rewritten, because the tool repeats whatever its documents say.

They asked for something that did not work. Mine is the AI output I rejected in week 3, the version that used the assignment's example as if it were a source. I found it by reading the target-audience line, not through any check in the tool. What I did about it is now a rule in the instruction file: cite only claims a person has accepted in the evidence log, and never general knowledge.

The questions on the interview guide and on field research exposed our weakest point. We could show how each internal sub-question became one to three open questions about real events, but no visit is booked and the consent text is not signed off, so nothing from an interview can

improve the tool yet.

What I change because of it: this week's page goes through the tool from search to check, so that the next page we show was made by the tool, with every claim on it accepted by a person.

WHAT DID MY BUILD ROLE PRODUCE?

The first running version of the tool, which is what we demonstrated on Monday. The build plan, this week's document, is not written yet on 2 October and is due on Monday. What I already know it changes for me: it asks for a test gate, a commit gate and a push gate, and our folder lives in a synced OneDrive folder with no commits at all. So the first subtask carries my name: moving the folder to a private GitHub repository, which also closes an open question in our PRD.

Week 5

AI use cases at SMEs

Session Mon 5 October 2026

WHAT DID THE TEAM PUBLISH?

Two handbook pages for the owner of a Dutch family-owned machine builder like Giesen: one on SME knowledge gaps and one on AI use cases at SMEs. They make one argument.

Page 1 finds that owners say they need concrete examples, while the evidence points to an earlier gap: 83% of Dutch SME manufacturers do not know which parts of their production suit AI, and 80% have underdeveloped data and sensor use. So the first thing an owner should find out is how reliable their own delivery dates are.

Page 2 finds that most firms use AI in the office rather than the factory, and that we could not find a single measured case of AI planning at a Dutch SME machine builder. The closest documented result, a peer-reviewed study of two German build-to-order firms, showed that predicting the delivery date at the quote lowered the risk of late delivery but carried 35% more error than the planners. So the page recommends a small paper test on the next ten quotes rather than buying a tool.

Both pages are on our GitHub and go on the course site once one reader outside the team has checked them for jargon. We also made our postcard from 2035 for China: a roaster builder in Wuhan whose AI proposes delivery dates while a planner decides.

WHAT DID I CONTRIBUTE TO IT?

Our interview has not happened, so I decided to go with plan B instead of waiting, and to have both pages say plainly what they cannot tell the owner because the firm itself was not asked. I did the tester's work this week: I checked 41 proposed claims against their saved sources, opened six of the sources myself, accepted 39 and rejected two. The two I rejected were survey figures quoted second-hand in an OECD blog post that turned out to be written by PayPal's head of government relations, a vendor citing a survey it is a partner in. I also checked the Dutch translations. Claude ran the searches and drafted the pages, the wiki and the postcard; I made the calls above, checked the evidence and pushed everything to our GitHub.

The moment that stuck with me: a search-engine summary said the German study had roughly halved the error compared with the planners. The paper says the opposite. If we had cited the summary, page 2 would have recommended the wrong thing. That is the whole case for keeping the exact source sentence with every claim.

WHAT DID THE GATES ASK, AND WHAT DID I CHANGE?

Our own platform pushed back on me twice. When I asked it to just produce the pages, it stopped at the tester's acceptance and at the wiki's go-ahead, because those are the two human checks our PRD promises. I changed the rules where they were ours to change: the assistant may now run several steps in a row, drafts may be written on proposed claims, and acceptance moved

to before publication. I also allowed published companies to be named, with written consent still required for the partner firm and its customers. I kept the two human checks, because each page has to say truthfully what the team checked.

The next gate is the first evaluation against our PRD, and one finding is already clear: until this week nobody recorded how long a session took, so our target of being at least 40% faster than by hand cannot be measured yet. I logged 20 minutes for this session, so the record starts now.

WHAT DID MY BUILD ROLE PRODUCE?

The knowledge architecture from this week's session, built into our own platform: an LLM wiki with a pause before it writes, source pages that record who wrote and who paid for each source and what was actually read, organisation pages, a Debates section, a log of every question asked, and a quotation check. I first accepted a restricted version, then changed my mind and asked for the complete pattern; the assistant added a firewall, which I kept: every wiki line is marked accepted or unchecked, and drafting may not take anything from the wiki.

Building it exposed a real weakness in our check step: it passed any sentence whose source had some accepted claim, so it now requires the specific claim. The run also showed limits I had not tested: two sites blocked scripted reading, ScienceDirect showed a bot check we deliberately did not get past, and the OECD's 2026 report could only be read as an abstract because our tools could not open the PDF. The wiki now holds 22 pages and its health check is clean.

THREE ECONOMIES: WHAT SURPRISED ME, AND DID I CHANGE MY MIND?

- **What surprised me:** how concrete China's plan is. Its 2026 manufacturing plan sets numbered targets for 2027: three to five general-purpose large models in manufacturing, 1,000 industrial AI agents and 100 industrial datasets. The front-runner we found, a roaster maker in Wuhan, could only show a vendor's claim. The state's targets were easier to verify than the company's AI.
- **Did I change my mind?** Partly. I started from the idea that Europe is simply behind. Making the postcard showed me the trade-off: in China the model and the pooled data come with the state's targets, while a Dutch builder keeps its own data and has the AI Act's protection but has to assemble the tools itself. Whether that counts as behind depends on who you think should own what the model learns.

WHAT I DON'T UNDERSTAND YET

Whether a machine builder that ships a handful of machines a month has enough order history for anything beyond a simple estimate. And whether our consent text will allow anonymised interview findings to go through the platform at all, which decides how plan A can feed the pages if the visit happens.